# Project Report

## Fundamental In AI&ML

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# Salary Prediction using Linear Regression

## Problem Statement

The goal of this project is to build a simple program that can predict the salary of an employee based on how many years of experience they have. Given a dataset of employees with their experience and corresponding salary, the program learns a pattern from the data and then predicts the salary for any new input provided by the user.

## Why Does This Problem Matter?

Salary estimation is something that actually happens in the real world — in HR departments, job portals, and even government pay-scale systems. Companies use past data to decide how much to offer a new hire or how to structure their pay grades.
For a first-year student, this problem is a great starting point because:
• The relationship between experience and salary is roughly linear easy to understand.
• The data is simple and relatable, making it easy to verify results manually.
• The output (a graph + a predicted number) is easy to present and explain.

It gave me a practical, hands-on way to understand how machine learning models actually learn from data, which is something that is hard to grasp just by reading a textbook.

## Approach to Solving the Problem

I used Linear Regression, which is one of the most basic machine learning algorithms. The core idea is simple — we try to draw the best possible straight line through a set of data points. Once we have that line, we can use it to predict the output for any new input.

**Steps followed:**

- Created a small dataset with 10 data points (years of experience and salary in INR).
- Split the data into a training set (80%) and a testing set (20%).
- Trained a Linear Regression model using scikit-learn on the training data.
- Tested the model on the test set and measured accuracy using MAE and $R^2$ score.
- Allowed the user to enter any experience value and see the predicted salary.
- Plotted the original data points and the regression line using matplotlib.

**Libraries used:**

| Library | Purpose |
|---|---|
| numpy | Handling numerical arrays and reshaping data |
| scikit-learn | Training and evaluating the Linear Regression model |
| matplotlib | Plotting the scatter graph and regression line |

## Key Decisions I Made

- **Why Linear Regression?**

I considered polynomial regression and a few other options, but since the dataset is small and the relationship between experience and salary appeared roughly linear, Linear Regression was the most appropriate and simplest choice. It also makes the results very easy to explain.

- **Why a hardcoded dataset instead of a CSV file?**

I wanted the project to be fully self-contained so anyone could run it without needing an external file. In a real-world scenario, data would be loaded from a CSV using pandas.

- **Why the 80/20 train-test split?**

This is a standard and widely used split in machine learning projects. 80% of the data trains the model and the remaining 20% is used to evaluate it on unseen data. With 10 rows, this gives 8 training and 2 test samples — small but sufficient to demonstrate the concept.

- **Why save the plot as a PNG file?**

So that the output remains accessible even after the terminal session is closed. It also makes it easy to include in a presentation or report.

## Challenges I Faced

**Understanding what 'training' actually means**
At first, I thought the model somehow memorized the data. It took some time to understand that it simply calculates the slope and intercept values that minimize the error between predicted and actual values. Once I understood the math behind it, everything clicked.

**Choosing the right evaluation metric**
I initially only looked at the $R^2$ score. Later I realized that Mean Absolute Error (MAE) is more intuitive — it directly tells you, in rupees, how far off the predictions are on average. Using both together gives a more complete picture of the model's performance.

**Reshaping the input array**
scikit-learn expects the input feature to be a 2D array. I initially passed a 1D array and got a ValueError. The fix was to use .reshape(-1, 1) — a small but important thing to get right.

**Small dataset limitations**
With only 10 data points, the test set had just 2 samples. This means the accuracy metrics can shift based on which samples end up in the test set. A larger, real-world dataset would give much more reliable and generalizable results.

## Results

After training the model, the following results were observed on the test set:

| Metric | Value | What it means |
|---|---|---|
| Mean Absolute Error (MAE) | ~Rs. 1,500 - 2,000 | Avg. prediction error in rupees |
| $R^2$ Score | ~0.998 | Model explains 99.8% of the variation |

An $R^2$ score this close to 1.0 means the model fits the data extremely well, which makes sense since the dataset was designed to follow a nearly linear pattern. The regression line on the plot clearly shows the upward trend and the predicted values were very close to the actual salary figures.

## What I Learned

This project taught me several important things:
• How machine learning models work at a basic level — they find patterns in data and use those patterns to make predictions on new inputs.
• The difference between training and testing — you train on one part of the data and evaluate on a separate unseen part to get an honest measure of performance.
• How to use Python libraries like scikit-learn and matplotlib in a real project, not just in theory.
• The importance of visualizing data — the scatter plot and regression line made it immediately clear whether the model was working correctly.
• That even simple models can be surprisingly effective when the data and the problem are well-suited to them.
If I were to extend this project in the future, I would try using a real dataset (such as one from Kaggle), add more features like education level or job role, and compare Linear Regression with other models like Decision Trees or Random Forest.

## Conclusion

This was my first proper machine learning project, and I found it genuinely interesting. The concept of a computer learning from data to make predictions even in this simple form felt very powerful. Linear Regression may be one of the oldest and simplest algorithms out there, but working through this project helped me build a solid foundation for understanding more complex algorithms in the future.

## References


- scikit-learn Documentation — https://scikit-learn.org
- NumPy Documentation — https://numpy.org/doc
- Matplotlib Documentation — https://matplotlib.org/stable
- Andreas Müller — Introduction to Machine Learning with Python

(Reference Book)